

Marketplace, AI Catalog, i18n & Data — Build Reference (current)

What the marketplace / AI-catalog system does today and how to run it. All original content or authorized-affiliate only — no retailer catalog is copied.

Marketplace (Facebook-style)

- Buy with **points** (closed loop, 1 pt = 1¢ in the local currency) or **card** (+10% markup).
- Purchases branch on `listing.source: platform_catalog` (platform sells, AI fulfillment), `user` (member sells/ships, AI-managed escrow), `affiliate` (retailer sells; we return the link).
- **Card orders don't fulfill until payment is captured** (no giveaway if card charging is switched on before a processor is wired).
- Members can **relist owned items with no personal info** (`relistItem`, seller shows as “Get Goods Gratis Member”). Third-party sellers post via `createMarketplaceListing` / the third-party seller page.
- Grid has search + sort (price/newest) + category filter.

AI Catalog — template-once, clone-per-country

- `aiCatalogSeed` builds a country-agnostic **template** set of original products across the taxonomy, generating each **image once** on the serverless GPU, then **clones** templates into every country in `CATALOG_COUNTRIES` — same base image, **localized price** (points = 1¢ local) and country context.
- Populates per country first; then third-party sellers add listings.

Taxonomy — exceeds a large retailer

- `taxonomy.ts`: **40 departments → 905 subcategories → ~21,700 AI browse nodes** (`aiBrowseNodeExpand` generates ~24 original browse nodes per subcategory). `getTaxonomy` powers the **Categories** page: department → subcategory → browse node → “find the real thing” product search.
- `aiCategoryImages` spins up original category tile images on the serverless GPU, once each.

“Find the real thing” search

- Aggregated, sortable search across **Amazon, Google Shopping, eBay, and the country's largest local retailer** (Rakuten/Flipkart/Mercado Livre/Otto/Walmart/...). Amazon carries your affiliate tag when configured (FTC-disclosed). Full-screen AI search bar with sort + price filters + in-catalog matches.

Serverless GPU images (AWS)

- `IMAGE_PROVIDER=aws_bedrock` (zero infra, pay-per-image) or `aws_sagemaker` (your SDXL/FLUX endpoint, scale-to-zero). Kill switch `CATALOG_IMAGES_ENABLED`, per-run cap `CATALOG_IMAGES_MAX_PER_RUN`.
- See `backend/.env.example` and `COST-AND-DEVHOURS-LEVERS.md` for setup + cost control.

Internationalization

- **88 country markets** (currency, language, flag, FX, top retailer) in `catalog.ts`; **56 currencies and 24 languages** in the frontend locale system. Currency conversion + translation apply site-wide, including marketplace prices.
- **Country auto-detected by geo-IP** (`LocaleContext`): the **top banner flag** and (unless the user chose otherwise) currency + language follow the user's real country. Cached 24 h with a browser-locale fallback.

Data collection & the self-improving Claude model

- Global click/event capture (`uxTracker.js`) + survey-honesty analysis feed behavioral entities.
- `site-model.ts` compiles a living context the model reads; `optimizer.ts` proposes changes; `experiments.ts` A/B-tests them with customers before non-sensitive ones auto-apply and sensitive ones require admin approval. Per-user profiles drive recommendations + the assistant chatbot.
- Users can opt out of recording, export their data, and delete their account.

Turn it on (admin → Platform Settings, or env)

- `CATALOG_COUNTRIES` (start with `US`), `IMAGE_PROVIDER` , `AI_DAILY_SPEND_CAP_USD` , `CATALOG_LISTINGS_PER_COUNTRY` (start ~80), optional Amazon Associate credentials.
- Scheduled jobs: `daily-catalog-seed` , `daily-category-images` , `daily-browse-node-expand` .

Guardrails that stay on

- Age 18+, tax thresholds, points non-cashable until cash-out is enabled, card charging off until a processor is wired, prohibited-item blocking, compliance keys excluded from AI optimization.